

COMSATS UNIVERSITY ISLAMABAD

ATTOCK CAMPUS



DEPARTMENT OF COMPUTER SCIENCE

COURSE	SQE Lab
INSTRUCTOR	Ms. Munazza
TASK	Assignment 4

Submitted By:

NAME	REGISTRATION NO.
JAVERIA IRUM	FA23-BSE-063
MARYAM FATIMA	FA23-BSE-007
AYESHA	FA23-BSE-032

Academic Task Tracker System

Introduction:

Academic Task Tracker System is a Python-based software application developed to help students manage their academic tasks. This system allows users to manage assignments, quizzes, mids, terminals, and project activities. Users can add tasks, view tasks, search tasks, update task status, and delete tasks. The system helps students stay organized and track their academic progress

Tools Used:

Python – Used for system development.

Visual Studio Code – Used for writing and running code.

SonarQube – Used for code quality analysis.

Snyk – Used for security analysis.

[GitHub](#) – Used for version control and project hosting.

Core Features:

1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit System

Manual Test Cases:

Test Case ID	Feature	Input	Expected Output	Actual Output	Status
TC-01	Add Task	PF, Assignment, Pending	Task added successfully	Task added successfully	Pass
TC-02	View Tasks	Press 2	Tasks displayed	Tasks displayed	Pass
TC-03	Search Task	PF	Task Found	Task Found	Pass

TC-04	Update Status	PF, Completed	Status updated	Status updated	Pass
TC-05	Delete Task	PF	Task deleted	Task deleted	Pass
TC-06	Exit	Press 6	Program closed	Program closed	Pass

Bug Report:

Bug 1

Bug ID: TASK_BUG_01

Bug Name: Tasks Disappear After Restart

Description:

After closing and restarting the program, previously added tasks are not available.

Steps to Reproduce:

1. Run the Academic Task Tracker system.
2. Add a new task.
3. Close the program.
4. Run the program again.
5. Select View Tasks.

Expected Result:

Previously saved tasks should remain available.

Actual Result:

Task list becomes empty.

Severity: High

Status: Fixed

Bug 2

Bug ID: TASK_BUG_02

Bug Name: Invalid Search Input

Description:

When a wrong course name is entered, the system cannot find the task.

Steps to Reproduce:

1. Run the system.
2. Select Search Task.
3. Enter a wrong course name.

Expected Result:

System should display "Task not found."

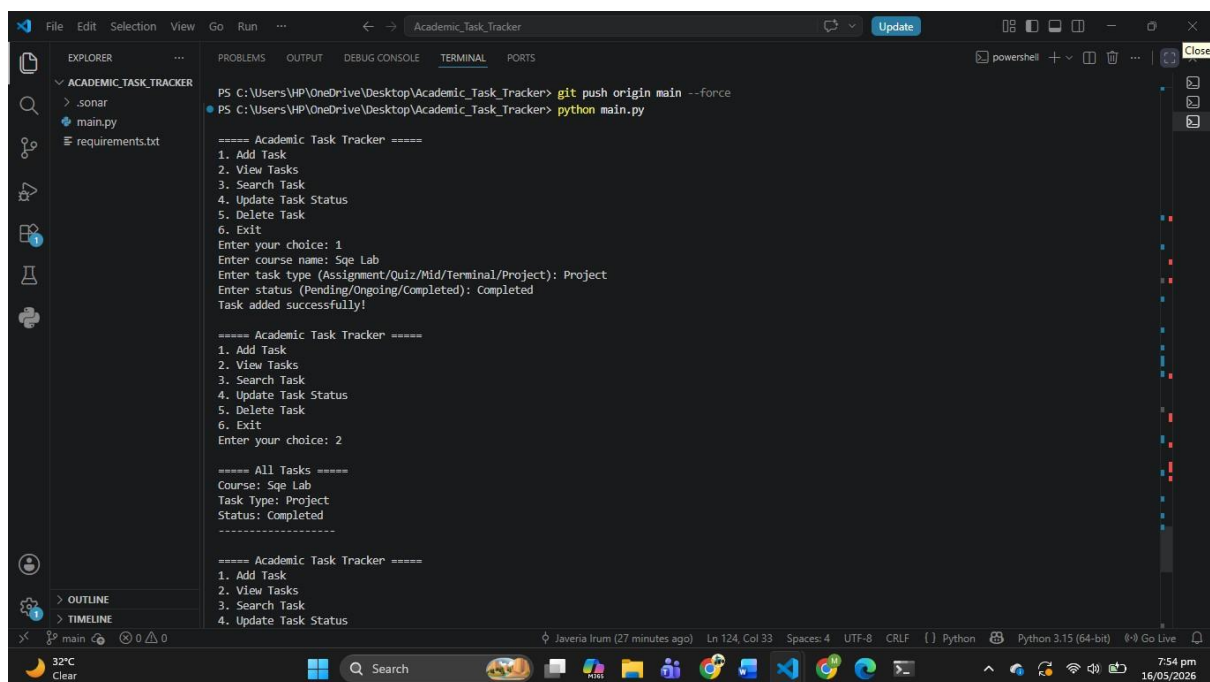
Actual Result:

Task not found message is displayed.

Severity: Medium

Status: Fixed

Program execution screenshots:



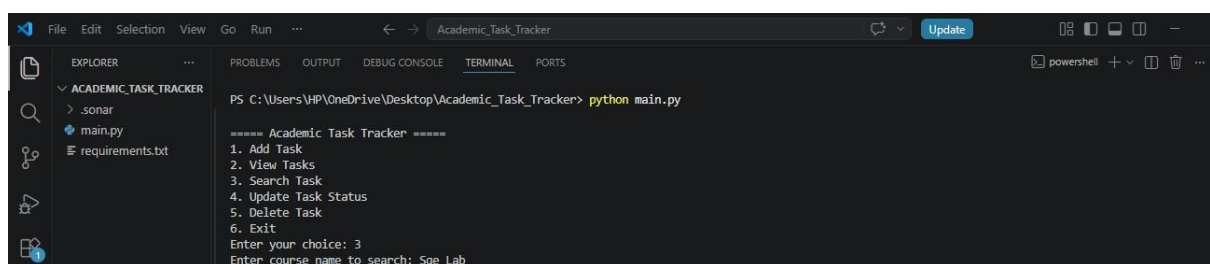
```
PS C:\Users\HP\OneDrive\Desktop\Academic_Task_Tracker> git push origin main --force
PS C:\Users\HP\OneDrive\Desktop\Academic_Task_Tracker> python main.py

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 1
Enter course name: Sqe Lab
Enter task type (Assignment/Quiz/Mid/Terminal/Project): Project
Enter status (Pending/Ongoing/Completed): Completed
Task added successfully!

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 2

===== All Tasks =====
Courses: Sqe Lab
Task Type: Project
Status: Completed
=====

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
```



```
PS C:\Users\HP\OneDrive\Desktop\Academic_Task_Tracker> python main.py

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 3
Enter course name to search: Sqe Lab
```



```
Task Type: Project
Status: Completed

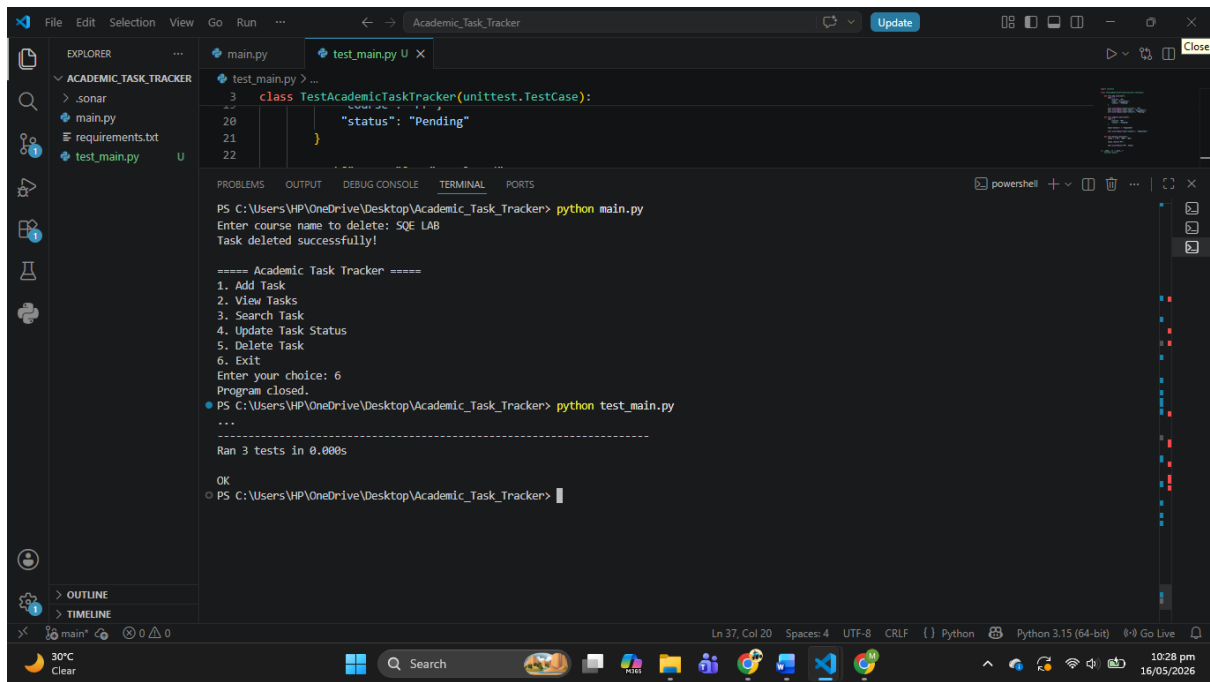
===== Academic Task Tracker =====
```

```
File Edit Selection View Go Run ... Academic_Task_Tracker Update
EXPLORER
ACADEMIC_TASK_TRACKER
  > .sonar
  > main.py
  requirements.txt
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\HP\OneDrive\Desktop\Academic_Task_Tracker> python main.py
Status: Completed

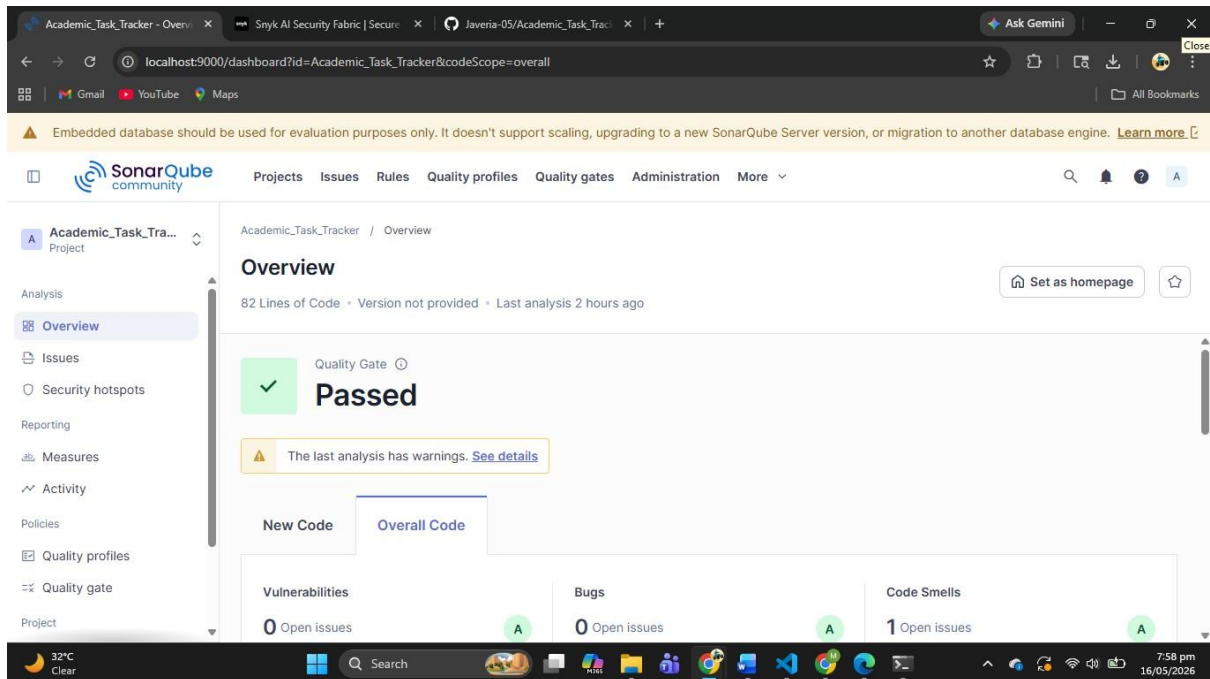
===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 4
Enter course name to update: SQE LAB
Enter new status: PROJECT IS COMPLETED
Status updated successfully!

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 5
Enter course name to delete: SQE LAB
Task deleted successfully!

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 6
Program closed.
PS C:\Users\HP\OneDrive\Desktop\Academic_Task_Tracker>
```



SonarQube analysis screenshots:



Academic_Task_Tracker - Overview | Snyk AI Security Fabric | Secure | Javeria-05/Academic_Task_Tracker | Ask Gemini

localhost:9000/dashboard?id=Academic_Task_Tracker&codeScope=overall

Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#)

SonarQube community

Projects Issues Rules Quality profiles Quality gates Administration More

Academic_Task_Tracker / Overview

Overview

82 Lines of Code • Version not provided • Last analysis 2 hours ago

Set as homepage

Vulnerabilities 0 Open issues A	Bugs 0 Open issues A	Code Smells 1 Open issues A
Accepted issues 0 Valid issues that were not fixed	Coverage 0.0% On 76 lines to cover.	Duplications 0.0% On 124 lines.
Security Hotspots 0 A		

32°C Clear 8:09 pm 16/05/2026

Academic_Task_Tracker - Issues | Snyk AI Security Fabric | Secure | Javeria-05/Academic_Task_Tracker | Ask Gemini

localhost:9000/project/issues?issueStatuses=OPEN%2CCONFIRMED&types=CODE_SMELL&tid=Academic_Task_Tracker

Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#)

SonarQube community

Projects Issues Rules Quality profiles Quality gates Administration More

Academic_Task_Tracker / Issues

Issues

My Issues All

Bulk Change Select issues Navigate to issue 1 issues 6min effort

main.py

Define a constant instead of duplicating this literal "Task not found." 3 times.

Code Smell Critical design

Open Not assigned L57 • 6min effort • 2 hours ago

1 of 1 shown

Community Build

32°C Clear 8:09 pm 16/05/2026

SonarQube Analysis Findings

Code was analyzed using SonarQube.

Issue 1: Code Smell

Repeated strings were found in the source code.

Suggested Fix:

Use constants or variables instead of writing the same text multiple times.

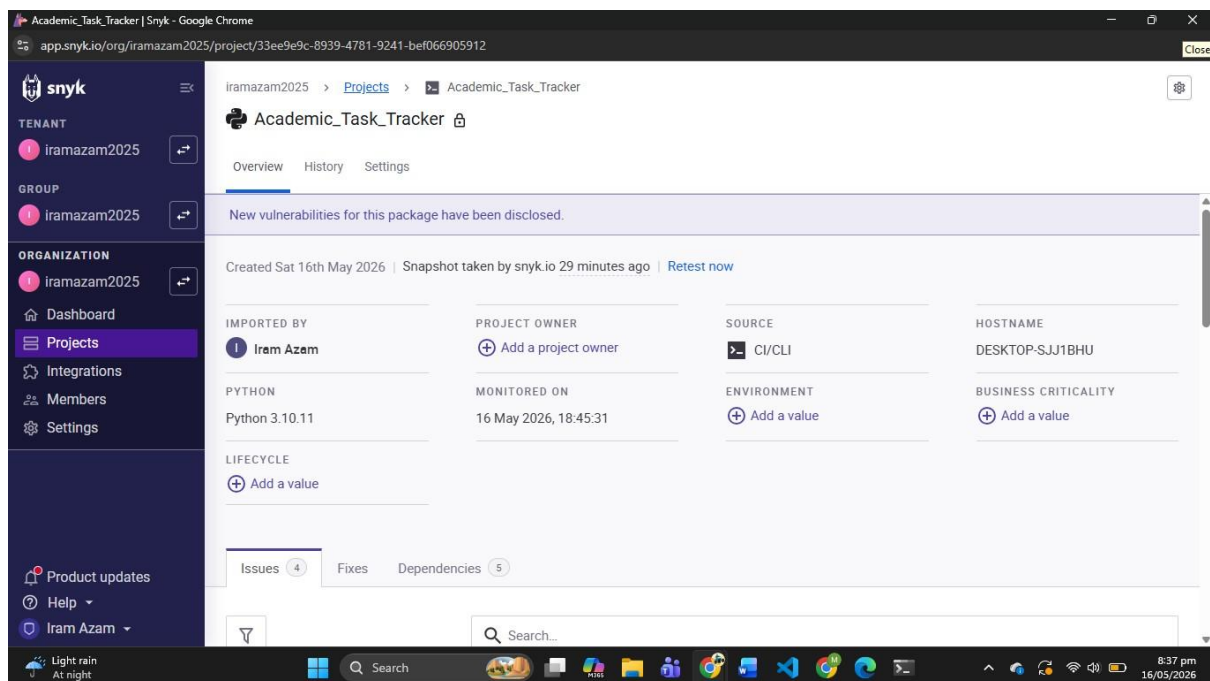
Issue 2: Maintainability Issue

Some code blocks were repetitive.

Suggested Fix:

Reusable functions should be used to improve code quality.

Snyk security scan screenshots:



Academic_Task_Tracker | Snyk - Google Chrome
app.snyk.io/org/iramazam2025/project/33ee9e9c-8939-4781-9241-bef066905912

snyk
TENANT
iramazam2025
GROUP
iramazam2025
ORGANIZATION
iramazam2025
Dashboard
Projects
Integrations
Members
Settings
Product updates
Help
Iram Azam

iramazam2025 > Projects > Academic_Task_Tracker

Academic_Task_Tracker

Overview History Settings

Issues (4) Fixes Dependencies (5)

Search...

4 of 4 issues Group by none Sort by highest priority score

H **urllib3 - Decompression Bomb** SCORE **731**
VULNERABILITY | ...

Introduced through requests@2.32.3 Exploit maturity NO KNOWN EXPLOIT
Fixed in urllib3@2.7.0

Show more detail

Ignore

SEVERITY
☐ Critical 0
☐ High 2
☐ Medium 2
☐ Low 0

PRIORITY SCORE
Scored between 0 - 1000

FIXED IN AVAILABLE
☐ Yes 4

30°C Clear 10:46 pm 16/05/2026

Academic_Task_Tracker | Snyk - Google Chrome
app.snyk.io/org/iramazam2025/project/33ee9e9c-8939-4781-9241-bef066905912

snyk
TENANT
iramazam2025
GROUP
iramazam2025
ORGANIZATION
iramazam2025
Dashboard
Projects
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Members
Settings
Product updates
Help
Iram Azam

iramazam2025 > Projects > Academic_Task_Tracker

Academic_Task_Tracker

Overview History Settings

☐ No 0

COMPUTED FIXABILITY
☐ Fixable 4
☐ Partially fixable 0
☐ No supported fix 0

EXPLOIT MATURITY
☐ Mature 0
☐ Proof of concept 1
☐ No known exploit 3
☐ No data 0

CVSS 4 EXPLOIT MATURITY
☐ Attacked 0

H **urllib3 - Insertion of Sensitive Information Into Sent Data** SCORE **696**
VULNERABILITY | ...

Introduced through requests@2.32.3 Exploit maturity NO KNOWN EXPLOIT
Fixed in urllib3@2.7.0

Show more detail

Ignore

M **requests - Insertion of Sensitive Information Into Sent Data** SCORE **606**
VULNERABILITY | ...

Introduced through requests@2.32.3 Exploit maturity PROOF OF CONCEPT
Fixed in requests@2.32.4

Show more detail

Rain coming 9:12 PM 8:37 pm 16/05/2026

Academic_Task_Tracker | Snyk - Google Chrome
app.snyk.io/org/iramazam2025/project/33ee9e9c-8939-4781-9241-bef066905912

snyk

TENANT
iramazam2025

GROUP
iramazam2025

ORGANIZATION
iramazam2025

Dashboard
Projects
Integrations
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Settings

Product updates
Help
Iram Azam

iramazam2025 > Projects > Academic_Task_Tracker

Academic_Task_Tracker

Overview History Settings

CVSS 4 EXPLOIT MATURITY

- Attacked 0
- Proof of concept 0
- Not defined 0

STATUS

- Open 4
- Patched 0
- Ignored 0

Introduced through requests@2.32.3
Fixed in requests@2.32.4
Show more detail

Exploit maturity PROOF OF CONCEPT

Ignore

M requests - Insecure Temporary File
VULNERABILITY | ... SCORE 419

Introduced through requests@2.32.3
Fixed in requests@2.33.0
Show more detail

Exploit maturity NO KNOWN EXPLOIT

Learn about this type of vulnerability

Ignore

30°C Clear 10:47 pm 16/05/2026

Academic_Task_Tracker | Snyk - Google Chrome
app.snyk.io/org/iramazam2025/project/33ee9e9c-8939-4781-9241-bef066905912

snyk

TENANT
iramazam2025

GROUP
iramazam2025

ORGANIZATION
iramazam2025

Dashboard
Projects
Integrations
Members
Settings

Product updates
Help
Iram Azam

iramazam2025 > Projects > Academic_Task_Tracker

Academic_Task_Tracker

Overview History Settings

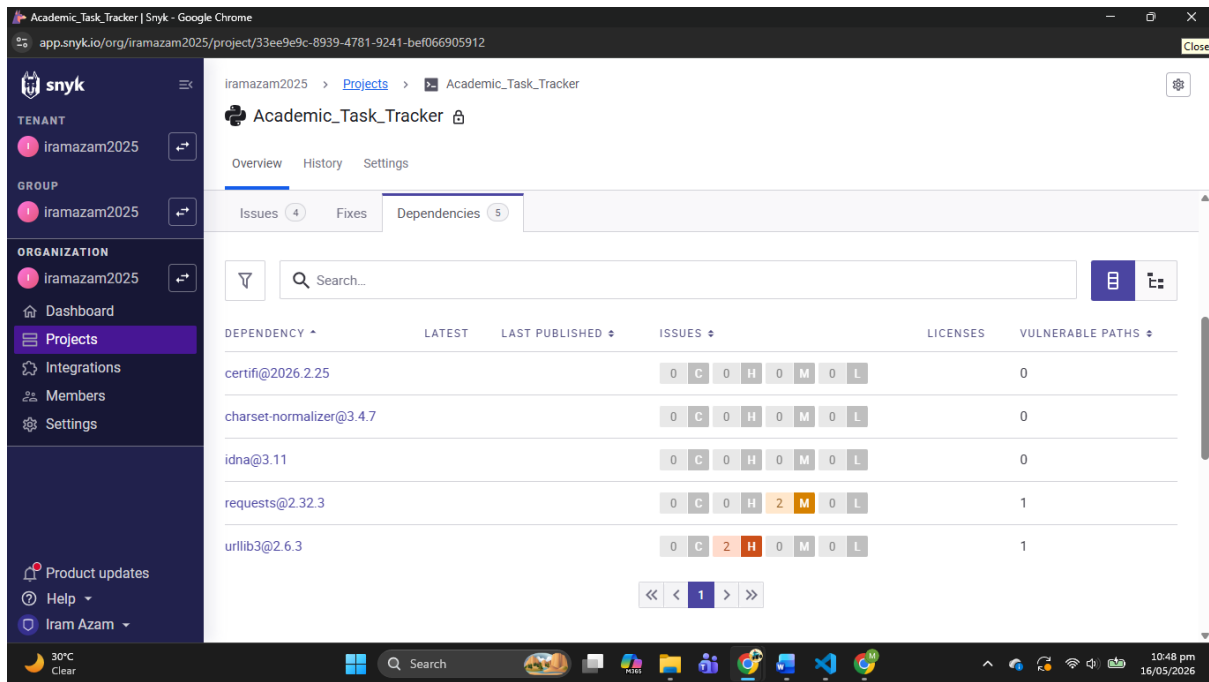
Issues (4) Fixes Dependencies (5)

Fix Advice

Upgradable issues (2)

- > Pin transitive dependency urllib3@2.6.3 to version urllib3@2.7.0
- > Upgrade direct dependency requests@2.32.3 to version requests@2.33.0

30°C Clear 10:47 pm 16/05/2026



Security Issues Found in Snyk

Issue 1: urllib3 – Decompression Bomb

This vulnerability may cause memory issues.

Fix:

Upgrade urllib3 to latest secure version.

Issue 2: urllib3 – Sensitive Information Exposure

Sensitive information may be exposed.

Fix:

Use updated secure package version.

Issue 3: requests – Sensitive Information Exposure

Data may become insecure during transmission.

Fix:

Upgrade requests package.

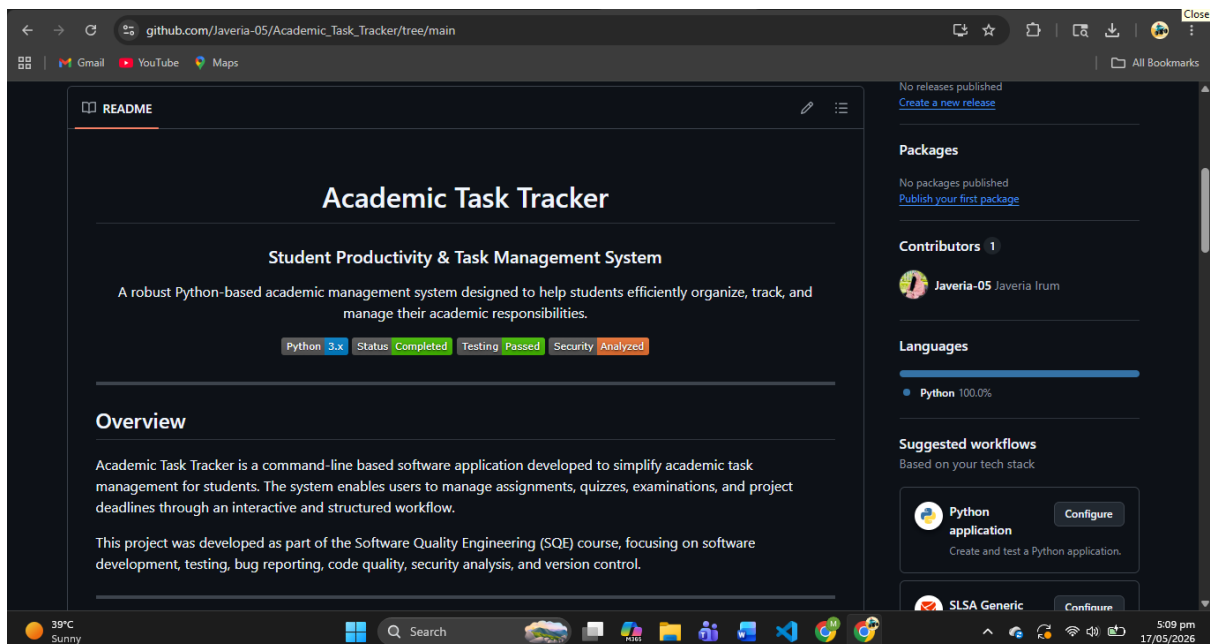
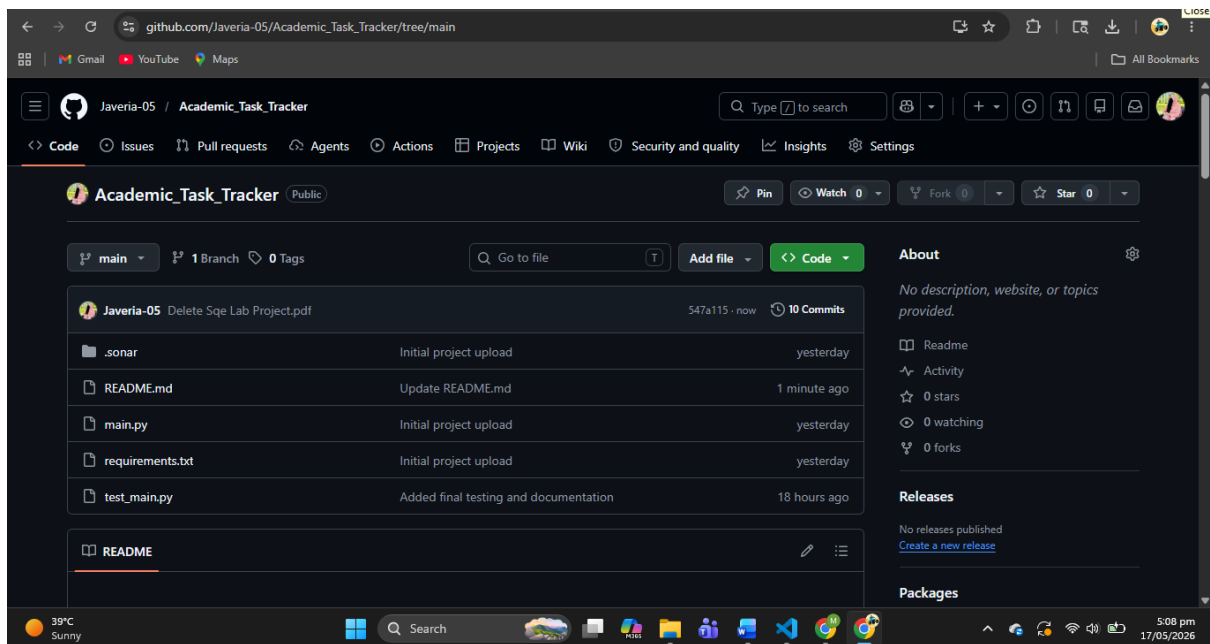
Issue 4: requests – Insecure Temporary File

Temporary files may be insecure.

Fix:

Use latest secure package version.

GitHub repository screenshots:



The screenshot shows a web browser window displaying the GitHub README for the repository 'Javeria-05/Academic_Task_Tracker'. The browser's address bar shows the URL 'github.com/Javeria-05/Academic_Task_Tracker/tree/main'. The README content is as follows:

README

Delete Task	Remove completed or unwanted tasks
Exit	Safely terminate the application

Technology Stack

Technology	Purpose
Python	Core System Development
VS Code	Development Environment
SonarQube	Static Code Analysis
Snyk	Security & Dependency Analysis
GitHub	Version Control

Software Quality Assurance

This project successfully implements:

